

A

MICRO PROJECT REPORT

on

“E-Commerce Sales & Customer Analysis”

Python for Data Science (BE05000231)

Submitted by

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Laxmi Institute of Technology-Sarigam
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October, 2026

CERTIFICATE

This is to certify that the Micro Project Report entitled “E-Commerce Sales & Customer Analysis” submitted by “Rahul Prasad” (Enrollment No.: 240860131044) of Semester 5th, Department of Computer Science & Engineering, Laxmi Institute of Technology-Sarigam, in partial fulfillment of the requirment subject Python for Data Science, has been satisfactorily completed during the Academic Year 2026-27. The work presented in this report is based on the novelty of the project, technical understanding, quality of report preparation, and project demonstration carried out during the allotted 8 Hours of Project Based Learning (PBL).

Date:

Place: LIT, Sarigam

Prof. Ankita V Rachh

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*This is to certify that the Micro Project Report entitled “**E-Commerce Sales & Customer Analysis**” submitted by “**Hitanshu Yadav**” (Enrollment No.: 240860131067) of Semester **5th**, Department of **Computer Science & Engineering**, Laxmi Institute of Technology-Sarigam, in partial fulfillment of the requirement subject **Python for Data Science**, has been satisfactorily completed during the Academic Year **2026-27**. The work presented in this report is based on the novelty of the project, technical understanding, quality of report preparation, and project demonstration carried out during the allotted 8 Hours of **Project Based Learning (PBL)**.*

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ACKNOWLEDGEMENT

We sincerely express our gratitude to **Laxmi Institute of Technology, Sarigam** for providing us the opportunity to carry out this micro project on “**E-Commerce Sales & Customer Analysis**” under the subject **Python for Data Science (BE05000231)** . We are thankful to our **Subject Incharge, Prof. Ankita V Rachh**, for valuable guidance, suggestions, and encouragement throughout the project planning and documentation process.

We also thank the **Head of the Department, Prof. Vrunda Patel** and all faculty members of the Department of Computer Science & Engineering for providing academic support and a suitable learning environment. We appreciate the cooperation of our team members in dividing responsibilities, discussing requirements, preparing schedules, identifying risks, and reviewing the project outcome.

This project helped us understand how a software idea can be converted into a planned project by defining scope, breaking work into manageable tasks, estimating resources and budget, scheduling activities, monitoring progress, and evaluating risks. We also learned the importance of communication, documentation, teamwork, and timely review. The experience has improved our understanding of practical project management and software project organization.

We are grateful to everyone who directly or indirectly supported us in completing this report.

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ABSTRACT

The “E-Commerce Sales & Customer Analysis” project focuses on analyzing e-commerce transaction data to identify meaningful patterns in sales performance, customer behavior, product performance, regional sales, shipping, and returns. The project uses Python, Pandas, and Matplotlib to perform data cleaning, transformation, analysis, and visualization. The dataset consists of 236 records and 12 original attributes. The data was cleaned by handling missing values, checking duplicate records, validating data types, and identifying inconsistent values. During validation, 8 price-mismatched records were identified. Data transformation was then performed by creating additional fields such as Revenue, Year, Month, Month Name, Total Amount, and Age Group. The analyzed results showed that Electronics was the best-performing category, Laptop was the highest-selling product, and the North region generated the highest sales. The 46–55 age group contributed the highest sales, while returned orders represented the largest shipping-status group. Headphones had the highest return rate among the products analyzed. Matplotlib was used to present the findings through various visualizations, including line charts and bar charts. The final insights were used to identify important business patterns and provide recommendations related to products, regions, customer segments, and returns. Overall, the project demonstrates how raw e-commerce data can be converted into useful business insights through a systematic data science workflow.

CHAPTER 1 : INTRODUCTION

E-commerce generates a large amount of sales and customer data that can be analyzed to understand business performance and customer behavior. Proper analysis of this data helps businesses identify high-performing products, profitable regions, customer purchasing patterns, and potential areas for improvement.

The “E-Commerce Sales & Customer Analysis” project uses Python-based data science techniques to analyze an e-commerce sales dataset. The project involves data cleaning, data transformation, data analysis, and data visualization using Python, Pandas, and Matplotlib.

The dataset contains information about customers, products, categories, prices, quantities, regions, shipping status, and order dates. The analysis is performed to extract meaningful insights from the raw data and present them through numerical results and visualizations.

The project also examines sales performance, product and category performance, regional sales, customer patterns, shipping status, and product return rates. The final insights can help identify business opportunities and support data-driven decision-making.

1.1 Project Scope

The project covers the following major stages:

Stage	Description
Data Cleaning	Handling missing values, checking duplicates, correcting data types, and validating numerical data
Data Transformation	Creating Revenue, Year, Month, Month Name, Total Amount, and Age Group
Data Analysis	Analyzing sales, products, categories, regions, customers, shipping, and returns
Data Visualization	Representing important findings using Matplotlib charts
Final Insights	Identifying important patterns and providing business recommendations

1.2 Background

The rapid growth of e-commerce has led to the generation of large amounts of customer and transaction data. This data contains valuable information about customers, products, sales, prices, quantities, regions, order dates, shipping status, and returns. Analyzing this information can help businesses understand their performance and customer purchasing behavior.

The “E-Commerce Sales & Customer Analysis” project focuses on analyzing an e-commerce dataset using Python and data science techniques. Pandas is used for data cleaning, transformation, and analysis, while Matplotlib is used to create visual representations of the results. The project follows a systematic process from raw data preparation to final insights and recommendations.

1.3 Need of the Project

Raw datasets may contain missing values, duplicate records, inappropriate data types, and inconsistent values, making direct analysis unreliable. Therefore, the data must first be cleaned and transformed before meaningful analysis can be performed.

This project is needed to:

- Understand the overall sales performance of the business.
- Identify the best-performing products and categories.
- Determine the highest-selling regions.
- Study customer purchasing patterns based on gender and age groups.
- Analyze different shipping statuses and product returns.
- Present important findings using clear and informative visualizations.
- Generate useful insights that can support data-driven business decisions.

1.4 Problem statement

E-commerce businesses collect large amounts of sales and customer data, but raw data does not directly provide meaningful information about business performance. Data-quality problems can also affect the reliability of the analysis.

Therefore, this project aims to clean, transform, analyze, and visualize e-commerce sales data to identify important patterns related to sales, products, categories, regions, customers, shipping, and returns.

CHAPTER 2 : OBJECTIVES , WORK FLOW

The main objective of the E-Commerce Sales & Customer Analysis project is to analyze the given e-commerce dataset and extract meaningful information about sales performance, customer behavior, products, regions, shipping, and returns.

2.1 Project Objectives

1. Data cleaning
To identify and handle missing values, duplicate records, incorrect data types, and inconsistent data.
2. Data transformation
To create useful fields such as Revenue, Year, Month, Month Name, Total Amount, and Age Group for further analysis.
3. Sales Analysis
To calculate and understand overall sales, quantity sold, average order value, and shipping fees.
4. Category Analysis
To identify the best-performing product categories based on sales, quantity, and number of orders.
5. Product Analysis
To determine which individual products generate the highest sales and have the highest demand.
6. Regional Analysis
To compare sales, quantities, and orders across different geographical regions.
7. Customer Analysis
To study purchasing patterns based on gender and age groups.
8. Shipping and Return Analysis
To analyze shipping statuses and identify products with higher return rates.
9. Data Visualization
To represent important findings using Matplotlib charts for understanding.

10. Final Insights and Recommendations

To identify important business patterns and provide recommendations based on the results of the analysis.

2.2 Work Flow

Data Cleaning → Data Transformation → Data Analysis → Data Visualization → Final Insights

The E-Commerce Sales & Customer Analysis project was completed by a team of three students. The project work was divided into different stages so that each member was responsible for a specific part of the data science workflow.

Team members

No.	Student Name	Enrollment No.	Assigned Work
1	Rahul Prasad	240860131044	Data Cleaning & Data Transformation
2	Aditya Sharma	240860131051	Data Analysis
3	Hitanshu Yadav	240860131067	Data Visualization & Final Insights

Rahul Prasad — Data Cleaning & Transformation

Responsible for preparing the raw dataset for analysis.

1. Data Cleaning

- Missing values
- Duplicate records
- Incorrect data types
- Invalid/inconsistent values
- Handling unknown values

2. Data Transformation

- Convert Order Date
- Create Year, Month, Month Name
- Create Revenue
- Create Total Amount
- Create age groups
- Validate calculated fields

Aditya Sharma — Data Analysis

Responsible for analyzing the cleaned and transformed dataset.

3. Data Analysis

- Sales summary
- Category analysis
- Product analysis
- Regional analysis
- Gender analysis
- Age-group analysis
- Shipping analysis
- Return analysis
- Return rate

Hitanshu Yadav — Data Visualization & Final Insights

Responsible for presenting the analysis through visualizations and extracting final conclusions.

4. Data Visualization

- Bar charts
- Line charts
- Pie/donut charts
- Regional comparison
- Product/category comparison
- Age-group analysis
- Shipping/return visualization
- Interactive/animated visualizations where useful

5. Final Insights

- Highest-selling region
- Best-performing products
- Best-performing categories
- Customer patterns
- Shipping/return patterns
- Business recommendations

CHAPTER 3 : DATASET DESCRIPTION

3.1 Dataset Overview

The dataset used in this project contains e-commerce transaction and customer information. It consists of 236 records and 12 original attributes. Each record represents an order and contains information about the customer, product, pricing, region, shipping, and order date.

3.2 Dataset Attributes

No.	Attribute	Description
1	Customer ID	Unique identification of the customer
2	Gender	Gender of the customer
3	Region	Region associated with the order
4	Age	Age of the customer
5	Product Name	Name of the purchased product
6	Category	Category of the product
7	Unit Price	Price of one unit of the product
8	Quantity	Number of units purchased
9	Shipping Fee	Shipping charge for the order
10	Shipping Status	Current status of the order
11	Order Date	Date on which the order was placed
12	Total Price	Total price of the purchased product(s)

3.3 Product Categories

The dataset contains three main product categories:

- Electronics
- Accessories
- Wearables

The products included in the dataset are:

Laptop, Smartphone, Monitor, Smartwatch, Headphones, Keyboard, and Mouse.

3.4 Dataset Format

The dataset is stored in CSV Link - [Drive File Link](#)

3.5 Tools and Technologies

Python

Python is the primary programming language used in this project. It is used for data processing, cleaning, transformation, analysis, and visualization.

Pandas

Pandas is a Python library used for working with structured data. In this project, it is used to:

- Load the CSV dataset.
- Handle missing values.
- Check duplicate records.
- Convert and validate data types.
- Perform data transformation.
- Group and analyze sales data.
- Calculate different statistics and metrics.
-

Matplotlib

Matplotlib is a Python visualization library used to represent the analysis graphically. It is used to create:

- Line charts
- Bar charts
- Sales comparisons
- Regional analysis charts
- Age-group analysis charts
- Shipping and return visualizations

CHAPTER 4 : DATA CLEANING AND TRANSFORMATION

4.1 Data Cleaning

Data cleaning was performed to improve the quality and reliability of the e-commerce dataset before analysis.

Python Code :

```
Import pandas as pd
Df = pd.read_csv("/storage/emulated/0/Download/E-commerce sales.csv")

# getting some info
Print(df.head())
Print(df.shape)
Print(df.info())
Print(df.isnull().sum())

# Handle missing values
Df["Region"] = df["Region"].fillna("Unknown")
Df["Age"] = df["Age"].fillna(df["Age"].median())
Df["Shipping Status"] = df["Shipping Status"].fillna("Unknown")

# Check missing values
Print("Missing values after cleaning:")
Print(df.isnull().sum())

# Checking duplicate rows
Print("Duplicate rows:")
Print(df.duplicated().sum())

Df["Order Date"] = pd.to_datetime(df["Order Date"], format="mixed")
Print(df.dtypes)

Print("Minimum Age:", df["Age"].min())
Print("Maximum Age:", df["Age"].max())
Print(df["Age"].describe())

Print("Minimum values:")

Print("Unit Price:", df["Unit Price"].min())
Print("Quantity:", df["Quantity"].min())
Print("Total Price:", df["Total Price"].min())
Print("Shipping Fee:", df["Shipping Fee"].min())
Df["Calculated Total"] = df["Unit Price"] * df["Quantity"]
Df["Price Difference"] = (
    Df["Calculated Total"] - df["Total Price"]
)
```

Observation

The given code successfully performs the initial data cleaning and validation of the e-commerce dataset.

- The dataset was loaded from a CSV file using Pandas.
- The initial structure, shape, data types, and missing values were examined.
- Missing values in Region and Shipping Status were replaced with "Unknown", while missing Age values were replaced using the median age.
- After cleaning, no missing values remained in the dataset.
- Duplicate records were checked, and 0 duplicate rows were found.
- Order Date was converted into the appropriate datetime format for further date-based analysis.
- The age range was validated and found to be 18 to 69 years.
- Minimum values of important numerical fields were checked to identify invalid values.
- Total Price was validated against $\text{Unit Price} \times \text{Quantity}$.
- 228 records passed the price validation, while 8 records showed mismatches.
- The temporary validation columns were removed after the checking process.

Overall Observation

The cleaning process improved the quality, consistency, and reliability of the dataset and prepared it for the next stage, Data Transformation.

4.2 Data Transformation

After cleaning, the dataset was transformed to create additional fields required for analysis.

Python Code :

```
Df["Revenue"] = df["Total Price"]
Print(df[["Total Price", "Revenue"]].head())

Df["Year"] = df["Order Date"].dt.year
Print(df[["Order Date", "Year"]].head())

Df["Month"] = df["Order Date"].dt.month
Print(df[["Order Date", "Month"]].head())

Df["Month Name"] = df["Order Date"].dt.month_name()
Print(df[["Order Date", "Month", "Month Name"]].head())

Df["Total Amount"] = df["Total Price"] + df["Shipping Fee"]
Print(df[["Total Price",
          "Shipping Fee",
          "Total Amount"]].head())

Bins = [17, 25, 35, 45, 55, 65, 75]
```

```

Labels = [
    "18-25",
    "26-35",
    "36-45",
    "46-55",
    "56-65",
    "66-75"
]

Df["Age Group"] = pd.cut(
    Df["Age"],
    Bins=bins,
    Labels=labels
)

Print(df[["Age", "Age Group"]].head(10))

Print("Updated dataset:")
Print(df.head())

Print("\nColumns:")
Print(df.columns)

Print("\nData types:")
Print(df.dtypes)

```

Full Cleaning & Transformation Code : [Full Code](#)

Observation

The data transformation process successfully created additional fields required for further analysis.

- A Revenue column was created from the existing Total Price column.
- Year was extracted from Order Date for yearly analysis.
- Month and Month Name were created to support monthly sales analysis.
- Total Amount was calculated by adding Total Price and Shipping Fee.
- Customers were categorized into six Age Groups: 18–25, 26–35, 36–45, 46–55, 56–65, and 66–75.
- The updated dataset was checked to confirm that the newly created columns were successfully added.
- After transformation, the dataset contained 18 columns, including the newly created fields.
- The data types of the transformed columns were also verified.

Overall Observation

The transformation process converted the cleaned dataset into a more analysis-ready structure, enabling time-based, financial, and age-group analysis in the next stage.

CHAPTER 5 : DATA ANALYSIS

Data analysis was performed on the cleaned and transformed dataset to identify important patterns and trends in sales, products, categories, regions, customers, shipping, and returns. The analysis was carried out using Pandas.

5.1 Sales Summary

The overall sales performance was analyzed using key measures such as total sales, total quantity sold, average order value, shipping fees, and number of orders.

Python Code :

```
Print("==== SALES SUMMARY ====")

Total_sales = df["Total Price"].sum()
Total_quantity = df["Quantity"].sum()
Average_order_value = df["Total Price"].mean()
Total_shipping = df["Shipping Fee"].sum()
Number_of_orders = len(df)

Print("Total Sales:", total_sales)
Print("Total Quantity Sold:", total_quantity)
Print("Average Order Value:", average_order_value)
Print("Total Shipping Fees:", total_shipping)
Print("Number of Orders:", number_of_orders)
```

Observation:

The dataset contains 236 orders with total sales of ₹306,390 and 670 units sold.

Metric	Result
Total Sales	₹306,390
Total Quantity Sold	670
Average Order Value	₹1,298.26
Total Shipping Fees	₹2,850.16
Total Orders	236

5.2 Category Analysis

Sales were grouped according to product category to identify the best-performing category.

Python Code :

```
print("==== CATEGORY ANALYSIS ====")

sales_by_category = df.groupby("Category")["Total Price"].sum()
quantity_by_category = df.groupby("Category")["Quantity"].sum()
orders_by_category = df["Category"].value_counts()

print("\nSales by Category:")
print(sales_by_category)
```

```
print("\nQuantity by Category:")
print(quantity_by_category)

print("\nOrders by Category:")
print(orders_by_category)
```

Observation:

Electronics is the best-performing category, generating the majority of total sales.

Category	Sales
Electronics	₹275,500
Accessories	₹17,090
Wearables	₹13,800

5.3 Product Analysis

Sales were analyzed for individual products.

Python Code :

```
print("===== PRODUCT ANALYSIS =====")

sales_by_product = df.groupby("Product Name")["Total Price"].sum()
quantity_by_product = df.groupby("Product Name")["Quantity"].sum()
orders_by_product = df["Product Name"].value_counts()

print("\nSales by Product:")
print(sales_by_product)

print("\nQuantity by Product:")
print(quantity_by_product)

print("\nOrders by Product:")
print(orders_by_product)
```

Observation:

Laptop is the highest-selling product, followed by Smartphone.

Product	Sales
Laptop	₹154,500
Smartphone	₹90,400
Monitor	₹30,600
Other Products	Remaining sales

5.4 Regional Analysis

Sales were compared across different regions.

Python Code :

```

print("==== REGIONAL ANALYSIS =====")

sales_by_region = df.groupby("Region")["Total Price"].sum()
quantity_by_region = df.groupby("Region")["Quantity"].sum()
orders_by_region = df["Region"].value_counts()

print("\nSales by Region:")
print(sales_by_region)

print("\nQuantity by Region:")
print(quantity_by_region)

print("\nOrders by Region:")
print(orders_by_region)

```

Observation:

The North region generated the highest sales, while the Unknown region contributed the lowest sales.

Region	Sales
North	₹91,550
West	₹72,720
East	₹67,340
South	₹64,280
Unknown	₹10,500

5.5 Gender Analysis

Sales were compared based on customer gender.

Python Code :

```

Print("==== GENDER ANALYSIS =====")

Orders_by_gender = df["Gender"].value_counts()
Sales_by_gender = df.groupby("Gender")["Total Price"].sum()
Quantity_by_gender = df.groupby("Gender")["Quantity"].sum()
Aov_by_gender = df.groupby("Gender")["Total Price"].mean()

Print("\nOrders by Gender:")
Print(orders_by_gender)

Print("\nSales by Gender:")
Print(sales_by_gender)

```

Observation:

Female customers generated slightly higher sales than male customers.

Gender	Sales
Female	₹159,040
Male	₹147,350

5.6 Age-Group Analysis

Customers were grouped by age to understand purchasing patterns.

Python Code :

```
print("==== AGE GROUP ANALYSIS ====")

orders_by_age = df["Age Group"].value_counts().sort_index()
sales_by_age = df.groupby("Age Group", observed=True)["Total Price"].sum()
quantity_by_age = df.groupby("Age Group", observed=True)["Quantity"].sum()

print("\nOrders by Age Group:")
print(orders_by_age)

print("\nSales by Age Group:")
print(sales_by_age)

print("\nQuantity by Age Group:")
print(quantity_by_age)
```

Full Python Code : [Full Code](#)

Observation:

The 46–55 age group generated the highest sales, while the 66–75 age group generated the lowest.

5.7 Overall Analysis Observation

The analysis shows that Electronics and Laptops are the strongest contributors to sales, while the North region has the highest regional sales. Customers aged 46–55 represent the strongest sales segment. The high number of returned orders and the relatively high return rates of some products indicate areas that may require further investigation.

CHAPTER 6 : DATA VISUALIZATION

Data visualization was performed to present the results of the analysis in a clear and understandable graphical form. Matplotlib was used to create the visualizations. A consistent black background with blue graphs/bars was used throughout the project.

6.1 Monthly Sales Visualization

A line chart was used to show how sales varied across different months.



Observation:

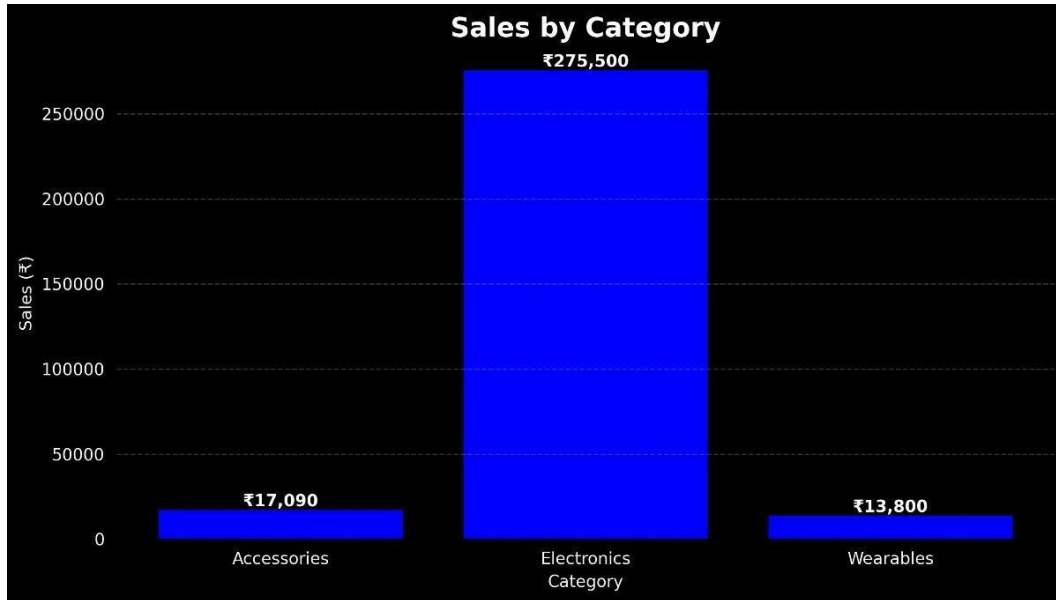
The monthly sales visualization helps identify changes and trends in sales over the year.

6.2 Sales by Category

A bar chart was used to compare sales among different product categories.

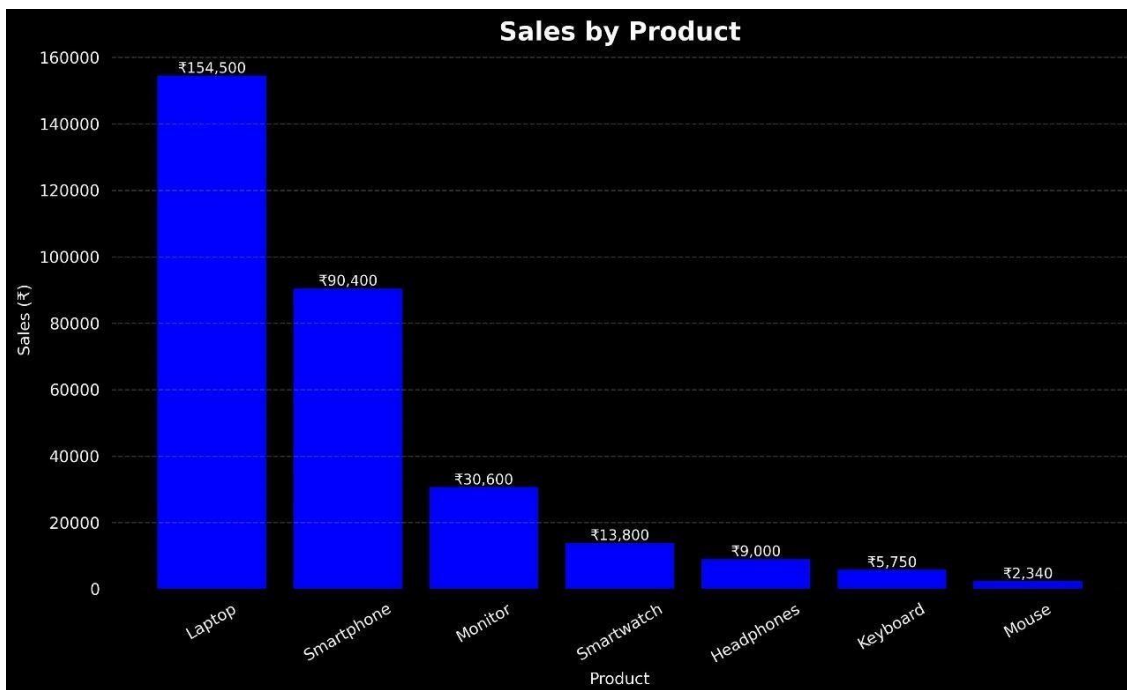
Observation:

The visualization clearly shows that Electronics is the dominant category in terms of sales.



6.3 Sales by Product

A bar chart was created to compare the sales generated by individual products.



Observation:

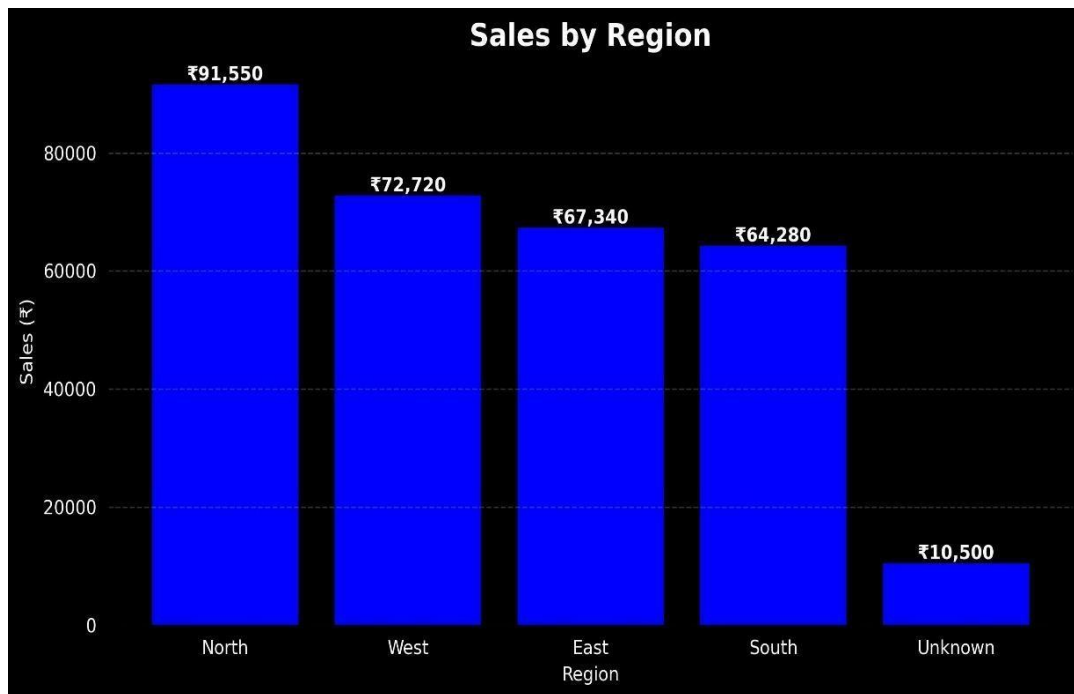
Laptop has the highest sales, followed by Smartphone.

6.4 Sales by Region

A bar chart was used to compare sales across different regions.

Observation:

The North region has the highest sales among the identified regions.



6.5 Sales by Gender

A bar chart was used to compare sales between male and female customers.

Observation:

Female customers generated slightly higher sales than male customers



6.6 Orders by Shipping Status

A bar chart was used to represent the number of orders according to shipping status.

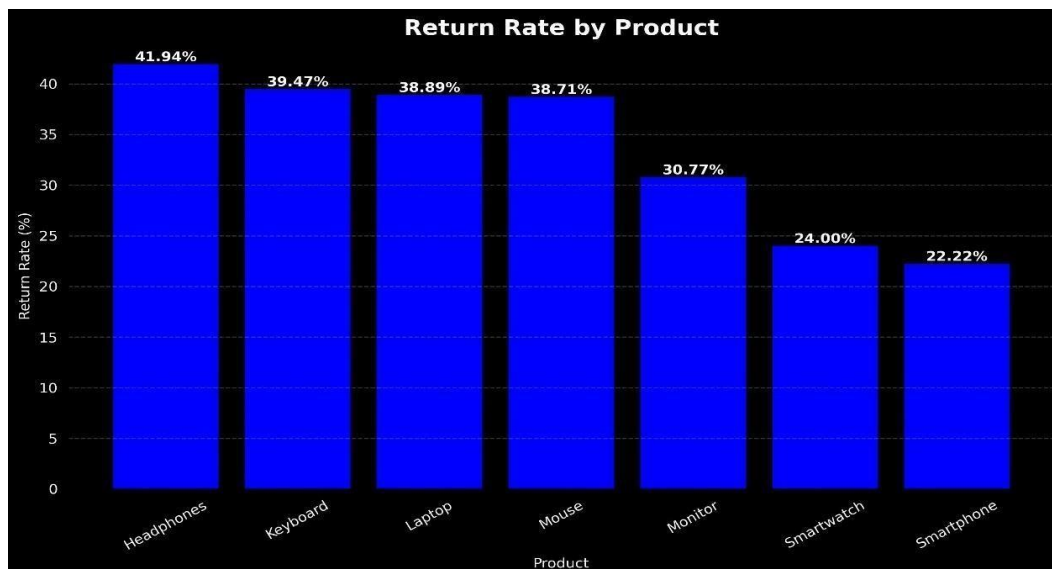


Observation:

Returned orders had the highest count among the shipping statuses.

6.7 Return Rate by Product

A bar chart was created to compare the percentage of returned orders for each product.



Python Code :

Import matplotlib.pyplot as plt

```
Monthly_sales = df.groupby("Month")["Total Price"].sum()
Plt.figure(figsize=(10, 5))
```

```
Ax = plt.gca()
Ax.set_facecolor("black")
Plt.gcf().set_facecolor("black")
Plt.plot(
    Monthly_sales.index,
    Monthly_sales.values,
    Color="blue",
    Marker="o",
    Linewidth=2
)
Plt.title(
    "Monthly Sales Analysis",
    Color="white",
    Fontsize=16,
    Fontweight="bold"
)
Plt.xlabel("Month", color="white")
Plt.ylabel("Sales (₹)", color="white")
Plt.xticks(range(1, 13), color="white")
Plt.yticks(color="white")
Plt.grid(True, linestyle="—", alpha=0.3)
Plt.tight_layout()
Plt.show()
```

```
Sales_by_category = df.groupby("Category")["Total Price"].sum()
Plt.figure(figsize=(9, 5))
Ax = plt.gca()
Ax.set_facecolor("black")
Plt.gcf().set_facecolor("black")
Plt.bar(
    Sales_by_category.index,
    Sales_by_category.values,
    Color="blue"
)
Plt.title(
    "Sales by Category",
    Color="white",
    Fontsize=16,
    Fontweight="bold"
)
Plt.xlabel("Category", color="white")
Plt.ylabel("Sales (₹)", color="white")
Plt.xticks(color="white")
Plt.yticks(color="white")
For I, value in enumerate(sales_by_category.values):
    Plt.text(
        I,
        Value,
        F"₹{value:,.}",
        Ha="center",
        Va="bottom" Color="white",
        Fontweight="bold"
    )
Plt.grid(
    Axis="y",
    Linestyle="— Alpha=0.3
)
Plt.tight_layout()
Plt.show()
```

For Full Code Tap here : [Full Visualization Code](#)

CHAPTER 7 : FINAL INSIGHTS AND BUISNESS RECOMMENDATION

Based on the data analysis and visualizations, the following key insights were identified.

7.1 Final Insights

Highest-Selling Region

- North is the highest-selling region with sales of ₹91,550.
- West is the second-highest-performing region with ₹72,720.

Best-Performing Products

- Laptop is the highest-selling product with ₹154,500 in sales.
- Smartphone is the second-highest-performing product with ₹90,400.

Best-Performing Category

- Electronics is the strongest category with sales of ₹275,500.
- Accessories and Wearables contribute comparatively smaller amounts.

Customer Patterns

- Female customers generated slightly higher sales than male customers.
- The 46–55 age group generated the highest sales at ₹108,390.
- The 56–65 age group was the next strongest contributor.

Shipping and Return Patterns

- Returned orders were the largest shipping-status group with 80 orders.
- Headphones had the highest product return rate at 41.94%.
- Smartphones had the lowest return rate at 22.22%.

7.2 Business Recommendations

Based on the findings, the following recommendations can be made:

1. Focus on high-performing products: Maintain sufficient inventory and marketing for products such as Laptops and Smartphones.
2. Strengthen the Electronics category: Since Electronics contributes the majority of sales, the business can continue expanding and promoting

products in this category.

3. Target the North region: Additional marketing and promotional activities can be considered in the North region because it generates the highest sales.
4. Focus on the 46–55 customer segment: Targeted offers and marketing campaigns can be designed for this high-performing age group.
5. Investigate high-return products: Products such as Headphones and Keyboards should be reviewed to understand the reasons for their relatively high return rates.
6. Improve return management: The high number of returned orders indicates a need to investigate product quality, product descriptions, customer expectations, and other possible causes of returns.

Overall Insight

The analysis indicates that Electronics, particularly Laptops, are the major contributors to sales, while the North region and 46–55 age group are important high-performing segments. At the same time, the high number of returns highlights an area where the business can improve its operations and customer experience.

CHAPTER 8 : RESULT AND DISCUSSION

8.1 Project Outcome

The project successfully completed the planned data science workflow for the e-commerce dataset. The raw dataset was cleaned, transformed, analyzed, and visualized using Python.

The final outcome includes:

- A cleaned dataset with 236 records and no remaining missing values.
- 0 duplicate records after cleaning.
- Identification of 8 price-mismatched records during validation.
- Creation of additional fields such as Revenue, Year, Month, Month Name, Total Amount, and Age Group.
- Analysis of sales, products, categories, regions, customers, shipping, and returns.
- Graphical representation of important findings using Matplotlib.
- Final business insights and recommendations based on the analysis.

8.2 Results

The major results obtained from the project are summarized below:

Analysis	Major Result
Total Sales	₹306,390
Total Quantity Sold	670
Total Orders	236
Highest-Selling Category	Electronics
Highest-Selling Product	Laptop
Highest-Selling Region	North
Highest-Sales Gender	Female
Highest-Sales Age Group	46–55
Highest Return Rate	Headphones — 41.94%
Most Common Shipping Status	Returned

These results provide an overall understanding of the sales and customer behavior present in the dataset.

8.3 Discussion

8.3.1 Strengths

- The project follows a complete data analysis workflow from cleaning to visualization.
- Data-quality issues were identified before performing analysis.
- Multiple aspects of the dataset were analyzed rather than focusing only on sales.
- Visualizations make the analytical results easier to understand.
- The project provides practical business recommendations based on the results.
- Python and Pandas make the analysis structured and reproducible.

8.3.2 Limitations

- The dataset contains only 236 records, so the results may not represent a large e-commerce business.
- The analysis is based on the available columns and does not include information such as customer reviews, payment methods, discounts, or marketing costs.
- The dataset contains 8 price-mismatched records, which may affect some financial calculations.
- The analysis is mainly descriptive and does not use advanced predictive or machine-learning models.
- The results are based on the available dataset and may change with a larger or different dataset.

8.3.3 Challenges

- During the project, several challenges were encountered:
- Handling missing values without losing useful records.
- Converting data into appropriate data types.
- Identifying inconsistent price values.
- Creating meaningful derived columns for analysis.
- Selecting appropriate visualizations for different types of data.
- Interpreting the results and converting them into useful business insights.

8.3.4 Lessons Learned

- Through this project, the team learned:
- The importance of data cleaning before analysis.
- How to use Pandas for real-world data manipulation.
- How to create new variables through data transformation.

- How grouping and aggregation can reveal business patterns.
- How Matplotlib can be used to communicate analytical results visually.
- How to interpret data and convert results into business recommendations.
- The importance of validating data instead of assuming that all recorded values are correct.

8.4 Overall Discussion

The project demonstrates that proper data preparation and analysis can turn raw e-commerce records into useful information. The results provide a clear view of sales performance, customer behavior, product performance, and return patterns while also highlighting areas that may require further investigation.

CHAPTER 9 : CONCLUSION

The E-Commerce Sales & Customer Analysis project successfully demonstrated the complete process of analyzing an e-commerce dataset using Python-based data science techniques.

The project began with data cleaning, where missing values, duplicate records, data types, unknown values, and inconsistent numerical values were examined and handled. The dataset was then transformed by creating useful fields such as Revenue, Year, Month, Month Name, Total Amount, and Age Group.

The cleaned and transformed data was analyzed to understand sales performance, product and category performance, regional sales, customer patterns, shipping status, and product returns. The results showed that Electronics was the strongest category, Laptop was the highest-selling product, and the North region generated the highest sales.

The analysis also identified important customer patterns, with the 46–55 age group contributing the highest sales. Shipping and return analysis showed that returned orders represented a significant portion of the dataset, with Headphones having the highest return rate.

Finally, the results were presented using Matplotlib visualizations, making the findings easier to understand and compare.

Overall, the project demonstrates how raw e-commerce data can be converted into meaningful insights through data cleaning, transformation, analysis, visualization, and interpretation. The findings can help businesses understand their performance and identify areas for improvement and future decision-making.

REFERENCE

The following resources were referred to during the development of the E-Commerce Sales & Customer Analysis project:

Python Documentation

Python Software Foundation. Python Documentation.

Used for Python programming concepts and functions.

<https://docs.python.org/3/>

Pandas Documentation

Pandas Development Team. Pandas Documentation.

Used for data cleaning, transformation, manipulation, and analysis.

<https://pandas.pydata.org/docs/>

Matplotlib Documentation

Matplotlib Development Team. Matplotlib Documentation.

Used for creating and customizing data visualizations, including bar charts and line charts.

<https://matplotlib.org/stable/>

Matplotlib

Pandas User Guide

Used as a reference for DataFrame operations, missing-value handling, grouping, time-series operations, and data processing.

Project Dataset

E-commerce Sales.csv — the dataset provided for the project and used for data cleaning, transformation, analysis, and visualization.

Kaggle <https://www.kaggle.comE>